



Driving-forces model on individual behavior in scenarios considering moving threat agents



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HIGHLIGHTS

- Driving-forces model is built considering scenarios including threat agents.
- An experiment is conducted to validate the key components of the model.
- The advanced Elliptical Specification II model is used for comparison.
- The fitting errors show the accuracy and stability of the Driving-forces model.
- The simulation results show the practicality of the Driving-forces model.

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ABSTRACT

The individual behavior model is a contributory factor to improve the accuracy of agent-based simulation in different scenarios. However, few studies have considered moving threat agents, which often occur in terrorist attacks caused by attackers with close-range weapons (e.g., sword, stick). At the same time, many existing behavior models lack validation from cases or experiments. This paper builds a new individual behavior model based on seven behavioral hypotheses. The driving-forces model is an extension of the classical social force model considering scenarios including moving threat agents. An experiment was conducted to validate the key components of the model. Then the model is compared with an advanced Elliptical Specification II social force model, by calculating the fitting errors between the simulated and experimental trajectories, and being applied to simulate a specific circumstance. Our results show that the driving-forces model reduced the fitting error by an average of 33.9% and the standard deviation by an average of 44.5%, which indicates the accuracy and stability of the model in the studied situation. The new driving-forces model could be used to simulate individual behavior when analyzing the risk of specific scenarios using agent-based simulation methods, such as risk analysis of close-range terrorist attacks in public places.

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1. Introduction

Recently, terrorists have created tension around the world. For example, the terrorist group Islamic State of Iraq and al-Sham (ISIS) and its followers carried out a series of attacks against France, England, and America in 2015. Governments across the world are facing great difficulty in allocating defensive resources to reduce the risk of terrorist attacks. Public

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places, especially those gathering large crowds of people, such as airports, train stations and theatres, have always been at a high risk of attack. For example, in the case of the November 2015 ISIS attack in Paris, 89 of the 130 victims were killed in the Bataclan theatre [1]. When an attack happens in a public place, it is difficult for individuals to get out of danger in time due to secondary effects, such as panic and overcrowding [2]. Thus, effective precautionary measures, such as evacuation signs and guides based on risk assessment of specific attack scenarios, become important in such situations.

The agent-based simulation (ABS) method is often used to simulate crowd behavior in different scenarios when conducting risk analysis. Some researches focused on crowd behavior in different layouts of circumstances under normal situations. Dai [3] simulated the pedestrian counter flow through bottlenecks. Johansson [4] simulated pedestrian behavior in waiting areas. More studies have focused on emergency situations, especially evacuation situations in different architecture layouts and scenarios, for example, the T-shaped intersection [5], the multi-room multi-floor building [6] and the escalator transfer area [7]. Some studies have also talked about key factors in emergency evacuation, such as the leadership effect [8] and the guidance effect [9]. Such studies even extended to terrorist attack scenarios. Wan [10] conducted a crowd evacuation simulation for bioterrorism in subway station environments.

In order to improve the accuracy of simulation results, building more suitable models and identifying the parameters for different types of agents in different scenarios becomes a critical point [11]. There are two categories of behavior models in general, macroscopic and microscopic models. Macroscopic models often treat crowd flow as a flowing continuum using the physical features of the gas flow or liquid flow [12]. Microscopic models, such as cellular automaton (CA) model [13] and social force (SF) model [14], consider an individual as a self driving particle. Among these models, the SF model is often used to set rules in ABS because of its good performance in describing individual behavior, especially the influences coming from other pedestrians, in continuous space [15].

The SF model was put forward by Helbing in 1995 [14]. The classical model considered other pedestrians and walls as influential factors to individual behavior. Then Helbing et al. [16] extended the model to escape panic situations, which considered the influence of a sliding friction force caused by body compression. Since then, many researchers have modified the details of the SF model, which made the model more suitable for different scenarios. For example, Kretz [17] modified the SF model accounting for oscillation situations. However, studies considering scenarios including threat agents, especially moving threat agents, such as an attacker in a terrorist attack scenario, are still inadequate. One reason may be that individual behavior in emergency situations is much more complex than normal situations. More factors need to be considered in such situations.

There still remain some aspects to be improved when looking at these related studies. Risk studies of crowds in different scenarios based on ABS methods often used the classical SF model as individuals' behavior rules without validating the accuracy of the model and the parameters in the specific scenario considered [18,7]. Some studies did not even list the values of the parameters in the social force model while doing simulations [5,10,19]. Studies focused on modifying the SF model under different conditions, especially under emergency situations, always lack validation from cases or experiments with real human subjects [20,4,9,19,8,17].

Due to these challenges, this study moves forward by building a driving-forces individual behavior model in scenarios including moving threat agents, and validating the key components of the model with experimental data. This could contribute to later ABS risk studies of emergency response. The remainder of this paper is structured as follows: Section 2 formulates the seven individual hypotheses and builds the driving-forces model; Section 3 introduces the experiment conducted to do the model calibration; Section 4 calibrates and validates the model; Section 5 presents the model fitting and simulation results, as well as the sensitivity analysis results; and Section 6 concludes and provides future research directions. Appendix A shows the flowchart of the data process programming; and Appendix B shows the probability density curves of the offset angle.

2. The model

In the basic SF model [16], the change of an individual's velocity $\mathbf{v}_i$ is influenced by the individual i 's desired speed v_i^0 and its direction $\mathbf{e}_i^0$, as well as the interaction forces generated by other pedestrians $\mathbf{f}_{ij}$ and walls $\mathbf{f}_{iW}$. The individual adapts his current speed to the desired speed in the period of characteristic time τ_i . The velocity change of individual i with weight m_i can be presented as Eq. (1).

$$m_i \frac{d\mathbf{v}_i}{dt} = m_i \frac{v_i^0 \mathbf{e}_i^0 - \mathbf{v}_i(t)}{\tau_i} + \sum_{j(\neq i)} \mathbf{f}_{ij} + \sum_W \mathbf{f}_{iW}. \quad (1)$$

Individual behavior during an emergency differs a lot from normal conditions. Firstly, most individuals in emergency situations do not think as rationally as they do in normal situations. So they behave more instinctively when they panic, and need more time to form a behavioral strategy. Secondly, the most important factor that influences individual behavior is the threat agent rather than other pedestrian or obstacle under emergency. Thirdly, most individuals run faster in emergency situations than usual. The running speed may not always remain constant. So there hardly exists a certain desired speed. Individuals keep adjusting their speed to fit the situation they face and to avoid becoming utterly exhausted. Finally, the existence of uncertainty and randomness makes an individual's behavior more difficult to predict when facing an emergency.

Based on these differences, seven individual behavioral hypotheses (H1–H7) are set considering the emergency situations including moving threat agents around the individual, as shown below:

- H1. There are three behavioral stages after noticing the threat agents: instinctive, strategic, and tactical stages.
- H2. Due to the threat agents, an individual would be driven by two types of forces. One leads in instinctive stage, called instinctive driving force. The other one leads in strategic and tactical stage, called strategic driving force.
- H3. (a) Instinctive driving force decreases with time and approaches zero; and (b) strategic driving force increases with time until a tactical strategy is formulated.
- H4. (a) Instinctive driving force increases when an individual is in an increased state of panic; and (b) strategic driving force increases when the magnitude of the threat's strength increases.
- H5. The threat agents within a certain range have same influence on an individual. Then the influence decreases with distance and approaches zero when the threat agent reaches the boundary of the individual's field of view.
- H6. (a) The individual is influenced by other people if there is physical contact; and (b) the individual would contemplate the situation and take action in order to avoid hitting the wall.
- H7. Inner resistance exists to keep an individual from utter exhaustion.

In this study, we only consider the threat agent whose speed is in the same order of magnitude as the individual, for example, an attacker with a sword in hand. Threat agent which moves much faster than human running speed is not considered, for example, a flying bullet.

The driving-forces model is built based on these behavioral hypotheses. According to hypothesis H2, the instinctive driving force $\mathbf{f}_{in}$ is generated when an individual feels unsafe. It is more timely. The strategic driving force $\mathbf{f}_{is}$ is generated because of an individual's willingness to get away from the threat. It incorporates both the effects of threat agents and the selected exit within a strategy. So it is more optimal than the instinctive driving force when leading the individual to get away from danger. There still remain forces from other individuals $\mathbf{f}_{ij}$ and walls $\mathbf{f}_{iw}$. We also consider the inner resistance $\mathbf{R}_i$ of the individual. The formulation of the driving-forces model is provided in Eq. (2). The details of each term will be introduced in Sections 2.1–2.4.

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_{in}(t) + \mathbf{f}_{is}(t) + \sum_{j(\neq i)} \mathbf{f}_{ij} + \sum_w \mathbf{f}_{iw} + \mathbf{R}_i. \quad (2)$$

2.1. Instinctive driving force

In the driving-forces model, the value of the instinctive driving force is decided by the maximum instinctive driving force $f_{i,\max}^n$, time factor $g^n(t)$ and panic degree factor $P_{i,C}^n$. The formulation of the instinctive driving force is shown in Eq. (3), and the set of inputs are introduced in the following paragraphs.

$$\mathbf{f}_{in} = f_{i,\max}^n \cdot g^n(t) \cdot P_{i,C}^n \cdot \mathbf{e}_i^n(t). \quad (3)$$

- The maximum instinctive driving force $f_{i,\max}^n$ controls the individual's maximum acceleration caused by the instinctive driving force. It is a personalized characteristic parameter, which means that different individuals may have different values of maximum instinctive driving force.
- The time factor $g^n(t)$ denotes the influence of time on the value of instinctive driving force, and has a range of $[0, 1]$. According to hypothesis H3(a), it slowly decreases with time and ends at time node t_n . That is, when $t < t_n$, we have $\frac{dg^n(t)}{dt} < 0$, $\frac{d^2g^n(t)}{dt^2} < 0$. When $t \geq t_n$, we have $g^n(t) = 0$.
- The panic degree factor $P_{i,C}^n$ denotes the influence of the panic's strength in condition C, according to H4(a). The panic degree factor has a range of $[0, 1]$.
- The direction vector $\mathbf{e}_i^n$ of the instinctive driving force has two types of expressions in two situations. First, if a certain exit E is selected by the individual, the direction vector would equal to the normalized vector $\mathbf{e}_{iE}$ pointing from the individual i to the exit E . Second, if there is no certain exit in the individual's field of view, then the direction vector would be decided by the normalized vector $\mathbf{e}_{ai}$ pointing from the threat agent to the individual, as well as a safe preference offset angle λ_i when an individual is not sure about where to go. The estimated value of offset angle will be introduced in Section 4.1. An indicator function X is defined (shown in Eq. (4)) before the direction vector is built in Eq. (5).

$$X = \begin{cases} 1, & \text{certain exit exists} \\ 0, & \text{certain exit does not exist} \end{cases} \quad (4)$$

$$\mathbf{e}_i^n = X \cdot \mathbf{e}_{iE} + (1 - X) \cdot \mathbf{e}_{ai}(t) \cdot \begin{pmatrix} \cos \lambda_i & -\sin \lambda_i \\ \sin \lambda_i & \cos \lambda_i \end{pmatrix}. \quad (5)$$

2.2. Strategic driving force

The strategic driving force leads the individual to get away from the threat tactically. It is decided by the maximum strategic driving force $f_{i,\max}^s$ and time factor $g^s(t)$. Also, the strategic driving force is decided by the threat strength $T_{i,a}^s$ and

distance affect $u_i^s(d_{ai})$ of the threat agent a . The influence of the exit is included if there is a certain exit E selected, and the weighing factor w_i^e is set to control the intensity. The formulation of strategic driving force is in Eq. (6), and the set of inputs are introduced in the following paragraphs.

$$\mathbf{f}_{is} = f_{i,\max}^s \cdot g^s(t) \cdot \left(\sum_a u_i(d_{ai}) \cdot T_{i,a}^s \cdot \mathbf{e}_{ai} + X \cdot w_i^e \cdot \mathbf{e}_{iE} \right). \quad (6)$$

- The maximum strategic driving force $f_{i,\max}^s$ controls the maximum acceleration caused by the strategic driving force, it is a personalized characteristic parameter.
- The time factor $g^s(t)$ denotes the influence of time on the value of strategic driving force, and has a range of [0, 1]. According to hypotheses H1 and H3(b), the strategic driving force generates within a decision making process, so we assume that $g^s(t)$ has an “s-curve” function form showing the different characteristics of the strategic driving force in three behavioral stages. In the instinctive stage, $g^s(t)$ is very small and increases slowly. In the strategical stage, $g^s(t)$ increases quickly. In the tactical stage $g^s(t)$ approaches the maximum value gradually after a tactical behavior strategy forms at time t_s .
- The distance factor $u_i(d_{ai})$ denotes the influence of distance d_{ai} between the threat agent and the individual, and has a range of [0, 1]. According to hypothesis H5, it equals to 1 when the distance is less than $d_{\min}$, then decreases with distance slowly, and ends when the distance equals to the $d_{\max}$, as shown in Eq. (7).
- The threat strength factor $T_{i,a}^s$ denotes the threat strength of agent a to the individual in condition C , according to hypothesis H4(b). It has a range of [0, 1], and is a personalized characteristic parameter.

$$u_i(d_{ai}) = \begin{cases} 1, & d_{ai} \leq d_{\min} \\ U_i(d_{ai}), & d_{\min} < d_{ai} \leq d_{\max} \\ 0, & d_{ai} > d_{\max}, \end{cases} \quad \frac{dU_i(d_{ai})}{dd_{ai}} < 0, \quad \frac{d^2U_i(d_{ai})}{dd_{ai}^2} < 0. \quad (7)$$

- The exit weighing factor w_i^e determines the influence intensity of the exit compared to that of the threat agent. For example, the weighing factor of an exit in the circumstance of a small room is larger than that of a big hall given the same conditions.

2.3. Repulsive force of other pedestrians and walls

When walking in normal situations, individuals tend to keep a distance from others. However, most individuals do not feel uncomfortable when they stay close with others in emergency situations. According to hypothesis H6(a), the repulsive force generates only if the individual has physical contact with other people. The repulsive force $\mathbf{f}_{ij}$ coming from another individual j is built upon the classical SF model [16], shown in Eq. (8). It includes the “body force” expression with the direction $\mathbf{e}_{ij}$ pointing from individual j to i and the “sliding friction force” expression with the tangential direction $\mathbf{t}_{ij}$ of the two individuals. The sliding friction force is decided by the tangential velocity difference $\Delta v_{ji}^t = (\mathbf{v}_j - \mathbf{v}_i) \cdot \mathbf{t}_{ij}$ of the two individuals.

$$\mathbf{f}_{ij} = k(r_{ij} - d_{ij})\mathbf{e}_{ij} + \kappa(r_{ij} - d_{ij})\Delta v_{ji}^t\mathbf{t}_{ij}, \quad r_{ij} > d_{ij} \quad (8)$$

where d_{ij} denotes the distance between centroids of individual i and j , and r_{ij} denotes the sum of the radii of the two individuals. And k and κ are constants that denote the intensity of the two forces.

In the classic SF model, the repulsive force of the wall increases exponentially with the distance, shown in Eq. (9) [16]. In this case, the individual receives a large acceleration in the normal direction when he gets close to the wall. This model cannot explain the phenomenon that an individual could walk along the wall. In our model, the individual contemplates the situation twice in each time step, according to hypothesis H6(b). The first time, the individual contemplates the trajectory generated by current states of motion. If the trajectory intersects with a wall, that wall would impact the individual. This impact is perpendicular to the wall, and only influence individual's normal component of the velocity towards the wall $\mathbf{v}_{in}$. In the second time of contemplation, the individual slows down with constant deceleration and stops at the side of the discussed wall. The value of deceleration is used to calculate the repulsive force of that wall, shown in Eq. (10). And d_{iw} is the horizontal distance from individual i to the wall w . Obstacles that cannot be passed through can also be treated as combination of walls.

$$\mathbf{f}_{iw} = A_w \exp(-d_{iw}/B_w) \quad (9)$$

$$\mathbf{f}_{iw} = \frac{m_i v_{in}^2}{2d_{iw}}. \quad (10)$$

2.4. Inner resistance

According to hypothesis H7, inner resistance exists in the driving-forces model due to the individual's physical exertion P , which is a function of speed. The inner resistance function $\mathbf{R}$ is the integration of physical exertion with time, pointing in



Fig. 1. (a) An experimental scene and (b) the starting locations of the attacker in twenty simulated attacks. (The middle point stands for the subject.)

the opposite direction of the individual's velocity, shown in Eq. (11).

$$\mathbf{R}_i = \int_t P_i(v_i) \cdot (-\mathbf{v}_i/v_i). \quad (11)$$

Physical exertion P grows rapidly with the speed of an individual; that is, $\frac{dP_i}{dv_i} > 0$, $\frac{d^2P_i}{dv_i^2} > 0$. It is a personalized characteristic parameter, and differs with heterogeneous classes of individuals. Older individuals who have larger physical exertion are more likely to feel tired than younger individuals after running for the same period of time.

3. Experiment

3.1. Introduction

We designed an experiment to investigate individual's reaction when facing a threat agent. The experiment was conducted on the platform of the gymnasium at Tsinghua University in Beijing, China on July 2nd–3rd, 2015. The threat agent was presented by a simulated attacker holding a sword in hand. Fig. 1(a) shows a scene in one experiment. The experiment's goal was to validate the influence of the key components in the driving-forces model, the instinctive driving force and the strategic driving force. In our experimental environment, there were no other pedestrians, walls, or certain exits, thus excluding the influence of other repulsive forces. Meanwhile, the subjects only needed to run for a couple of seconds, so the inner resistance could be negligible. For this preliminary study, there was only one moving threat agent in the individual's field of view simulated by a strong male student who did not know any of the subjects prior to the experiment. There were 43 students who participated, including 17 females and 26 males. The average age was 21 years with a standard deviation of 1.8 years. All subjects were right handed. The process of each simulated attack was conducted as follows:

- The subject was told the circumstances of the experiment and task.
- The subject stood on the center of measurement, closed his/her eyes, and put on the earphone with simulated background noise from public crowds.
- The simulated attacker stood on the starting location and prepared to attack.
- A cry for help was played into the earphone as a starting signal.
- The subject opened his/her eyes, and took action after noticing the attacker.
- The subject and the attacker stopped after a second signal.

For each subject, one simulated attack was repeated twenty times, with the attacker starting at twenty different locations in a random order. Twenty locations were arranged at eight directions: Front (F), Front Left (FL), Front Right (FR), Left (L), Right (R), Back Left (BL), Back Right (BR), and Back (B), with four initial distances between the attacker and the subject: 2, 4, 6, and 8 m. The red numbers in Fig. 1(b) show the starting location of the attacker in each of the twenty simulated attacks.

3.2. Data processing

Two forms of data were recorded after the experiment, including the locations of the subject's first steps and the videos of simulated attacks. The data processing contains three stages: data input, projective transformation, and calculation. The flowchart of data processing programming is in Appendix A. The coordinates of all points on the camera image are recorded during the data input stage. During the projective transformation stage, calibration of the four vertexes of the transformed

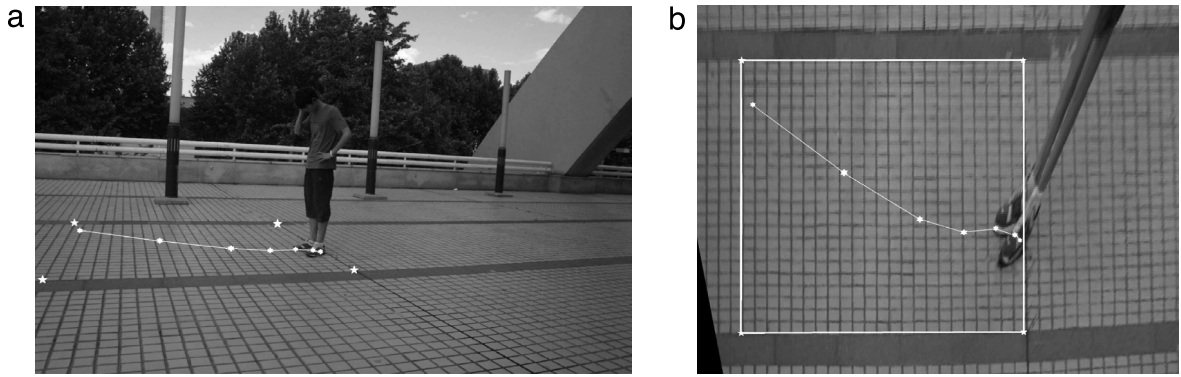


Fig. 2. (a) A camera image sample and (b) its projective image.

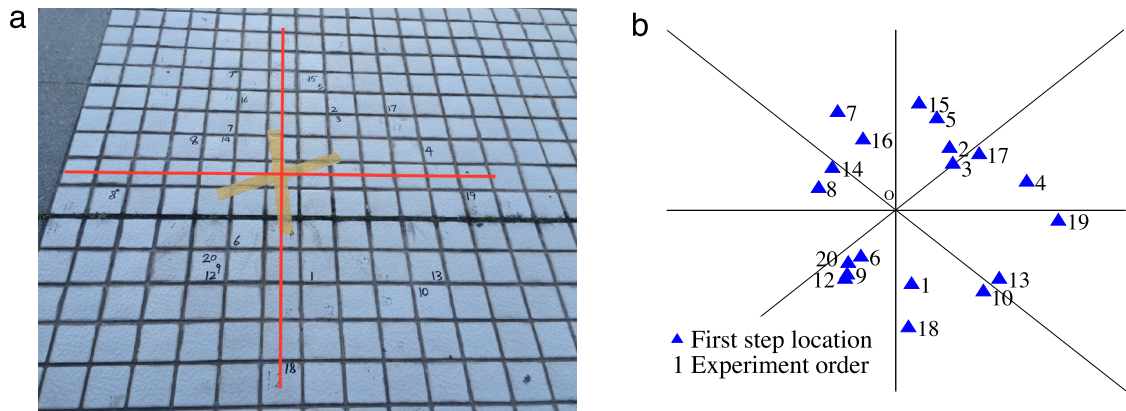


Fig. 3. One subject's first step (a) in photos and (b) in data format.

region is used to calculate the projection matrix transforming the camera image into the projective image. The projection matrix is a 3×3 matrix with eight parameters. Among these parameters, four refer to the rotation and scaling of the images before and after projective transformation, two refer to the translation and the other two refer to the projection. A camera image sample and its projective image are shown in Fig. 2(a), (b). On the projective image, the tiles in the transformed region change back to squares after transformation. The trajectory is proportional to x - y axes of the two-dimensional coordinate system on the floor. Then the actual size of the trajectory is obtained through the calibration. The life-size motion parameters of the individual are obtained during the calculation stage.

4. Model calibration and validation

In this section, data from the experiment introduced in Section 3 are used to calibrate the driving-forces model in Section 2. An advanced SF model is used to compare with the driving-forces model for validation.

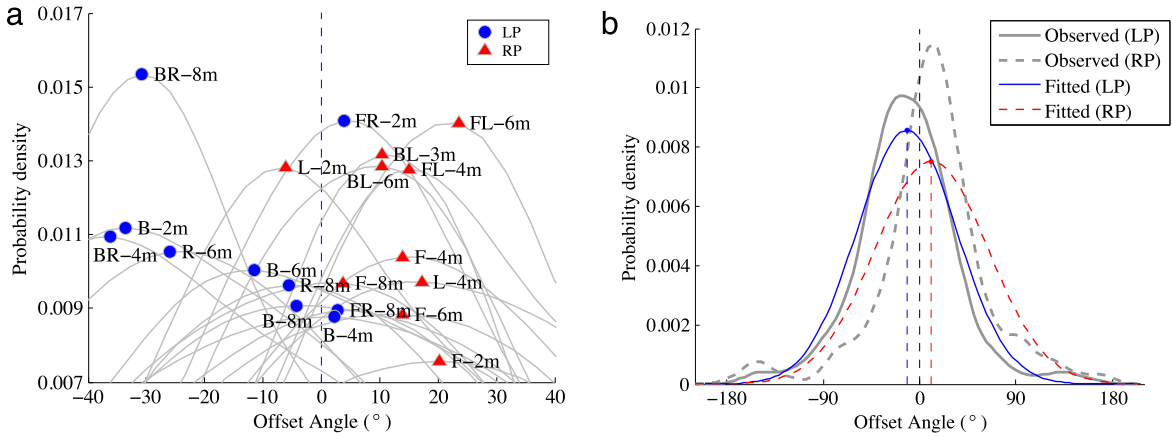
4.1. Description of offset angle

We recorded the locations of the subjects' first steps, shown in Fig. 3(a), (b). There are twenty sets of data where the attacker starts in twenty different locations. For each set, 43 data are collected from all the subjects. The offset angle λ_i stands for the angle difference between the subject's and the attacker's first steps. If the individual is rational and only has a repulsive force from the attacker, the expected running direction should be the same direction as that of the attacker. This means the offset angle should be in a symmetric distribution around zero.

Table 1 shows the mean value of the offset angle in each set. We use the normal distribution function to fit the probability density curves of the offset angle, shown in Appendix B. The data shows that the offset angles do not fluctuate around zero, but have different deviation values in different sets. For example, when the attack is 4 m away in front of the subject, the probability density curve of the offset angle from all subjects is close to a normal distribution (the p -value is 0.626 when doing the Shapiro–Wilk normality test [21], which is larger than the confidence level of 0.05), and has an average offset angle of 14.146° . That means, the running directions of most individuals are the back right, and has an average counterclockwise

Table 1Mean value of the offset angle ($\bar{\lambda}_i$).

Location	$\bar{\lambda}_i$ (deg)	Location	$\bar{\lambda}_i$ (deg)	Location	$\bar{\lambda}_i$ (deg)	Location	$\bar{\lambda}_i$ (deg)
F-2m	−6.829	F-4m	14.146	F-6m	10.250	F-8m	−4.634
FR-2m	0.500	FL-4m	26.098	FL-6m	−4.390	FR-8m	−5.125
L-2m	−10.610	L-4m	32.195	R-6m	−34.250	R-8m	−10.921
BL-2m	20.244	BR-4m	0.750	BL-6m	14.375	BR-8m	−1.585
B-2m	−15.366	B-4m	2.195	B-6m	−0.325	B-8m	5.250

**Fig. 4.** (a) The probability density curves of offset angle in twenty sets; (b) Distribution of left/right partition.

offset angle of 14.146° when the attacker is 4 m in front. The third line of Table 1 also shows that, the absolute value of offset angle is larger when the attacker is on left or right side of the subjects.

Fig. 4(a) shows the probability density curves of all twenty sets. The dots and triangles represent the value of offset angle with the highest probability in each set. It shows that when the attacker is in the Front, Front Left, Left and Back Left (the dots), the offset angle λ_i is more likely to be less than zero. And when the attacker is in the Front Right, Right, Back Right and Back (the triangles), the offset angle are more likely to be larger than zero. According to this, the value of the offset angle is separated into two categories based on the location of the attacker. The left partition (LP) includes the Front side and the right partition (RP) includes the Back side. The standard deviation confidence interval is $[-11.9^\circ \pm 46.6^\circ]$ for the left partition, and $[11.1^\circ \pm 53.0^\circ]$ for the right partition, as shown in Fig. 4(b). Individual's running forward preference when the threat is in the directions except the dead ahead or dead astern could be the cause of the offset angle. In addition, when the threat is dead ahead, individuals prefer to run into the right back, and when the threat is dead astern, it would be the front left. At the same time, the offset angle is a personalized characteristic parameter, which means that different individuals may have different offset angle values. However, due to the limitation of the experiment, it cannot be deduced that the results also fit for the left-handed individuals.

4.2. Assumptions of initial condition

Some assumptions about the initial conditions are made before simulations. All these functional forms meet the behavioral hypotheses and the model settings specified in Sections 2.1 and 2.2. The parameter baseline values are set according to the common sense. And then the sensitivity analyses of parameters are analyzed in Section 5.2.

- The instinctive driving force time factor $g^n(t)$.

$$g^n(t) = \begin{cases} 1, & t = 0 \\ 1 - t^2/t_n^2, & 0 < t \leq t_n \\ 0, & t > t_n \end{cases}; \quad t_n = 0.5 \text{ s}. \quad (12)$$

- The strategic driving force time factor $g^s(t)$.

$$g^s(t) = \begin{cases} 0, & t = 0 \\ -0.5 \cos\left(\frac{\pi}{t_s} \cdot t\right) + 0.5, & 0 < t \leq t_s \\ 1, & t > t_s \end{cases}; \quad t_s = 1.5 \text{ s}. \quad (13)$$

- The strategic driving force distance factor $u_i(d_{ai})$.

$$u_i(d_{ai}) = \begin{cases} 1, & d_{ai} \leq d_{\min} \\ U_i(d_{ai}), & d_{\min} < d_{ai} \leq d_{\max} \\ 0, & d_{ai} > d_{\max} \end{cases}; \quad U_i(d_{ai}) = -\frac{(d_{ai} - d_{\min})^2}{(d_{\max} - d_{\min})^2} + 1;$$

where $d_{\min} = 2 \text{ m}$; $d_{\max} = 12 \text{ m}$. (14)

4.3. Calibration method

Genetic algorithm is often used to calibrate individual behavior model; that is, using both the current simulated and experimental data to calculate the next point on the simulated trajectory when doing the simulation [22,23]. However, a complete simulated trajectory cannot be obtained through this method because the motion parameters of the individual analyzed is reset according to the experimental data in each time interval. To make the simulated results less dependable on the experimental data, the genetic algorithm is not used to do the model calibration. In this paper, the trajectory is simulated after assigning the initial values and the ranges to the personalized characteristic parameters. Then the parameters are determined through tree traversal algorithm for the best fitting results.

The trajectory obtained from the experiment is discrete. It has a time interval of 0.25 s, which is also used as the time interval of simulation. The fitting error function F considers the differences between the point sets on the trajectories obtained from experiment $\{\mathbf{r}_i(t_k)\}$ and that from the simulation $\{\mathbf{r}'_i(t_k)\}$, while k is the order of the points on the trajectory. The formula of the fitting error function is shown in Eq. (15).

$$F = \frac{1}{k} \sum_k \frac{\|\mathbf{r}'_i(t_{k+1}) - \mathbf{r}_i(t_{k+1})\|}{\|\mathbf{r}_i(t_{k+1}) - \mathbf{r}_i(t_k)\|}. \quad (15)$$

The ranges of the offset angle in the two partitions for each individual are set from the calculated confidence interval analyzed in Section 4.1. The maximum instinctive driving force and maximum strategic driving force are set in the range of $[0, 10] \text{ ms}^2 \times \text{m}_i$.

To validate the driving-forces model, it is compared with the Elliptical Specification II (ESII) model, which is an advanced SF model put forward by Johansson et al. in 2008 [22]. The difference between the ESII model and the classical Circular Specification model [16] is that the ESII model considers the relative velocity of other pedestrians and uses the concept of “repulsive potential” ($V_{ij}(b)$) to calculate the repulsive force. Thus the pedestrians standing on the elliptical equipotential line are influenced equally by the individual standing on one of the foci of the elliptical. The model uses the semi-minor axis length of the elliptical equipotential line b_{ij} to replace the distance variable in the classical Circular Specification SF model. The repulsive potential is calculated by Eq. (16) using the exponential form of b_{ij} , where b_{ij} is calculated by the characteristics of an ellipse, shown in Eq. (17). The repulsive force is got by the gradient of the repulsive potential, shown in Eq. (18). Lastly, the change of velocity can be calculated by the classical form of SF model, shown in Eq. (19).

$$V_{ij}(b) = A_i \exp(-b_{ij}/B_i) \quad (16)$$

$$2b_{ij} = \sqrt{(\|\mathbf{d}_{ij}\| + \|\mathbf{d}_{ij} - v_j \Delta t \mathbf{e}_j\|)^2 - (v_j \Delta t)^2} \quad (17)$$

$$\mathbf{f}_{ij}(\mathbf{d}_{ij}) = -\nabla_{\mathbf{d}_{ij}} V_{ij}(b_{ij}) \quad (18)$$

$$m_i \frac{d\mathbf{v}_i}{dt} = m_i \frac{v_i^0 \mathbf{e}_i^0 - \mathbf{v}_i(t)}{\tau_i} + \mathbf{f}_{ij} + \mathbf{f}_{iw}. \quad (19)$$

To use this model, the attacker’s influence is treated similarly to the influence from other pedestrian in the ESII model using Eqs. (16)–(19). The desired speed v_i refers to experimental data. In order to make the ESII model more adaptable, the direction of the desired speed $\mathbf{e}_i^0$ in the ESII model is set as the direction $\mathbf{e}_{ai}$ from the attacker to the individual with a range of offset angle λ_i analyzed in Section 4.1. This is an advanced deformation of the original model. Otherwise, the individual will run directly into the same direction of the attacker, which completely deviates from reality. The calibration procedure of the ESII model consists of three steps. First, the greatest speed of the individual in each simulated attack is used to set the desired speed, and the mean value of the offset angle is used to set the direction of the desired speed. Second, parameters A_i and B_i are optimized. For each subject, these two parameters should remain constant in all simulated attacks. Third, the optimized values of A_i and B_i are used to determine the exact desired speed, including its direction in each simulated attack.

5. Results

The fitting error function built in Section 4.3 is used as the comparison measurement of the trajectories generated by the driving-forces model and the advanced ESII model. Then the two models are applied to simulate the individual behavior in a single room including an attacker.

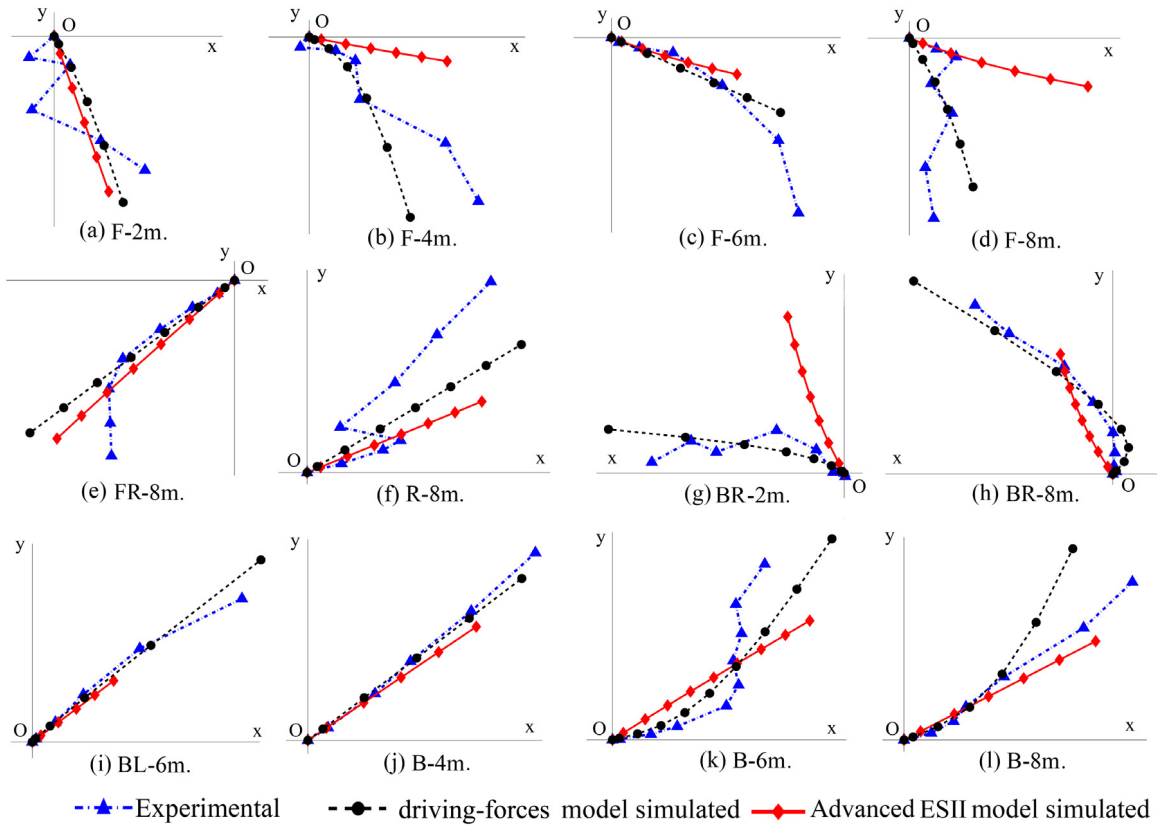


Fig. 5. Comparison of simulated and experimental trajectories of (a) F-2m, (b) F-4m, (c) F-6m, (d) F-8m, (e) FR-8m, (f) R-8m, (g) BL-2M, (h) BR-8m, (i) BL-6m, (j) B-4m, (k) B-6m, (l) B-8m.

5.1. Fitting results

The transformation data from seven subjects with high-quality video records are used to do the calibration. The fitting errors of the simulated results generated by the driving-forces model and the advanced ESII model are shown in Table 2, as well as the maximum instinctive driving force and maximum strategic driving force of each subject generated by the driving-forces model. The standard deviation confidence interval of fitting error is calculated using the data from all simulated attacks of one subject. Table 2 shows that the results generated by the driving-forces model have a smaller fitting error. It has reduced the fitting error by an average of 33.90% when compared with the advanced ESII model, which indicates better accuracy. Meanwhile, the standard deviation of the fitting error reduced by 44.50% on average, indicating the greater stability of the proposed driving-forces model. The optimized parameters A_i and B_i of the advanced ESII model are also found to be close to zero for all subjects, indicating that the force generated by the attacker does not match the exponential form with the distance between the attacker and the individual, as hypothesis H5 supposed. In contrast, other pedestrians cause discomfort only if they are close enough to the individual. This influence decreases rapidly when the distance increases.

To clarify, the comparisons of trajectories simulated by the driving-forces model and the advanced ESII model with the experimental trajectories of one subject are shown in Fig. 5. Because of the location of the camera in the experiment, only 12 trajectories were obtained from this subject. Fig. 5 shows that there are more inflection points on the experimental trajectories than on the simulated trajectories, presenting the larger randomness in the running path of the individual. Also, the experimental trajectory seldom appears as a straight line (Fig. 5(a), (b), (d), (g)), which shows the difficulty for an individual to run into a certain direction in the studies situation. This indirectly supports hypothesis H2. At the same time, compared to the advanced ESII model, trajectories generated by the driving-forces model are closer to the experimental trajectories in both temporal and spatial dimensions (Fig. 5(b), (c), (d), (f), (i), (j), (k)). Specifically, when the individual makes a big change in the running direction, the advantage of driving-forces model is more significant (Fig. 5(h)).

5.2. Sensitivity analysis

The box plots of panic degree factor $P_{i,C}^n$ under different distances between the attacker and the subject are shown in Fig. 6. It shows that the value of panic degree factor is almost the same when the distance is 2 m or 4 m, but decreases when

Table 2

The fitting error of two models, the maximum instinctive driving force and the maximum strategic driving force.

Subject No.	F (driving-forces model)	F (ESII model)	$f_{i,\max}^n$ (N)	$f_{i,\max}^s$ (N)
1	1.50 ± 0.48	2.49 ± 1.28	117.0	182.0
2	1.31 ± 0.30	2.30 ± 0.81	163.2	380.8
3	0.93 ± 0.25	1.78 ± 0.88	220.8	420.0
4	1.14 ± 0.47	1.43 ± 0.38	252.0	201.6
5	1.27 ± 0.28	2.01 ± 0.49	138.0	216.0
6	0.95 ± 0.19	1.39 ± 0.47	196.0	268.8
7	0.82 ± 0.09	1.00 ± 0.14	155.0	300.0

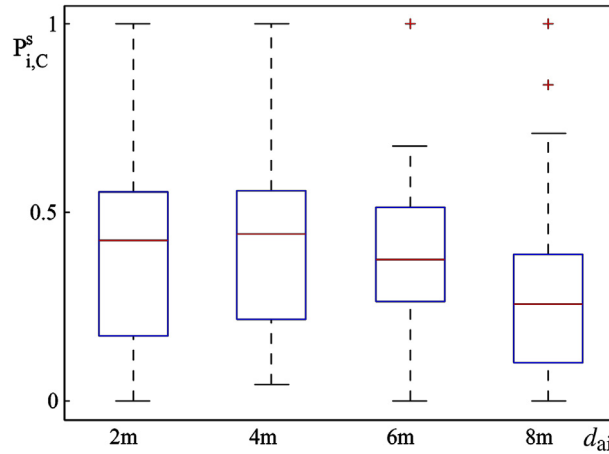


Fig. 6. Box plots of the panic degree $P_{i,C}^s$ under different distances.

the distance is greater than 4 m. This fits the hypothesis H4(a), because the subject would feel a great deal of panic when the attacker is close by, and then the panic reduces when the distance increases.

To validate the baseline values of parameters t_n , t_s , $d_{\min}$, and $d_{\max}$ set in Section 4.2, the simulations are repeated using different values of these four parameters. The fitting errors between the simulation results and the experimental data in different values of the parameters are calculated. The red lines in the box plots show the medians of the fitting errors after removing the outlier data. The smaller it is, the better fit. Fig. 7(a), (b), (c), (d) suggest: (a) the duration of the instinctive driving force t_n is 0.5 s; (b) the duration of making a behavioral decision t_s is 1.5 s; (c) the threat agents within 2–3 m have almost the same influences to the individual; and (d) when the distance $d_{\max}$ is over 12 m, the direct effects of the threat agents disappear. These indicate that the baseline values of parameters set in Section 4.2 are reasonable.

5.3. Simulation results

The driving-forces model and the advanced ESII model are applied to simulate individual behavior in an indoor circumstance including an attacker. The room is a 10 m long and 10 m wide square. The exit is 3 m wide. There is a 4 m long and 2 m wide rectangular obstacle in the room, which does not block the individual's view of the exit. The attacker runs after the individual at a speed of 3 m/s, and the navigation mesh method is used to aid the attacker in pathfinding. The parameters of the driving-forces model are set as follows: $m_i = 80$ kg, $f_{i,\max}^n = 400$ N, $f_{i,\max}^s = 500$ N, $P_{i,C}^n = 0.8$, $T_{i,a}^s = 0.8$, and $w_i^e = 0.8$. The parameters of the advanced ESII model are set as follows: $m_i = 80$ kg, $v_i^0 = 5$ m/s, $\tau_i = 0.5$ s, $A_i = 2 \times 10^3$ N, $B_i = 0.08$ m, $A_w = 3 \times 10^3$ N, and $B_w = 0.2$ m. Also, the physical exertion function is added to each of the two models, ensuring that the velocities of the individuals are within a reasonable range. The functional form is assumed to be $P_i(v_i) = A_p v_i^{B_p}$. The baseline parameter values are set as follows: $A = 0.15$ kg/ms, and $B = 2$. The assumptions satisfy the property of the physical exertion function (Eq. (11)) introduced in Section 2.4.

The simulated trajectories of the two models are shown in Fig. 8. The circles stand for the tracepoints of the individual, and the triangles stand for that of the attacker. The time step of the simulation is 0.15 s. In order to show the relative locations of the attacker and the individual at a time more clearly, every four points on the trajectories are marked in dark color. Fig. 8 shows that, the driving-forces model simulates the individual's avoidance of the attacker and the obstacle better than the advanced ESII model. For example, the individual would not "rebound" off the wall like the situation generated by the advanced ESII model. The maximum value of repulsive force of the wall f_{iw} produced by the advanced ESII model in the simulation is 1492.3 N. It is unreasonably high and gives the individual an acceleration of 18.7 m/s², which is beyond the affordability of any individual. By comparison, the maximum value of repulsive force of the wall produced by the driving

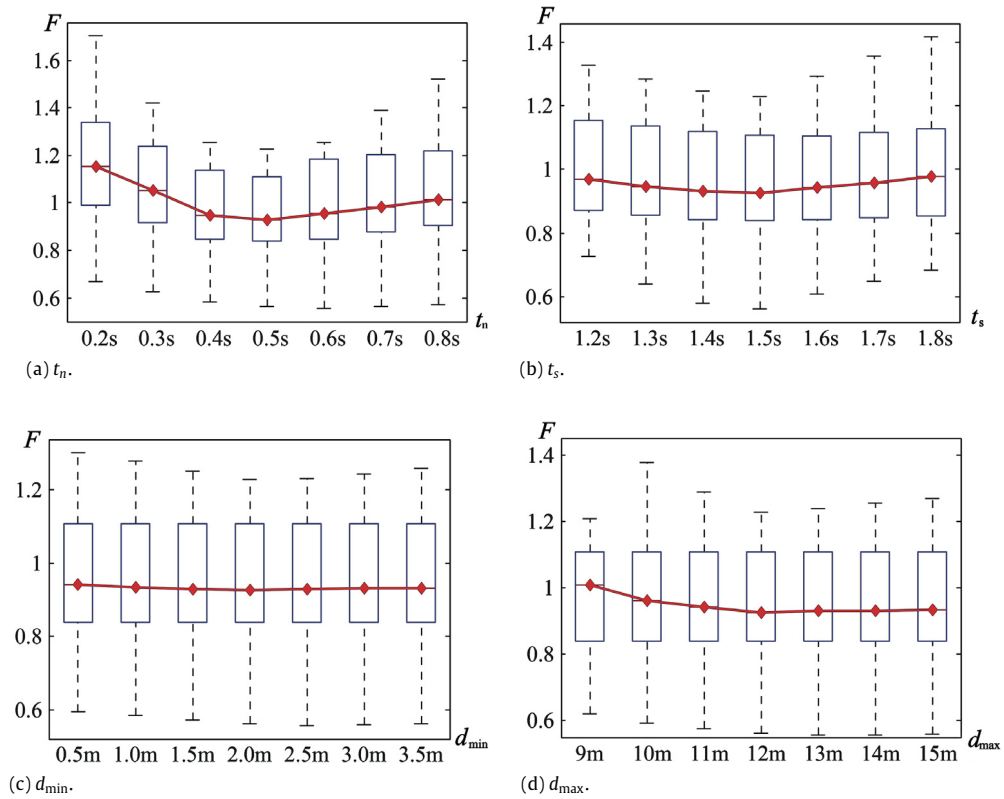


Fig. 7. Box plot of fitting with different parameter values.

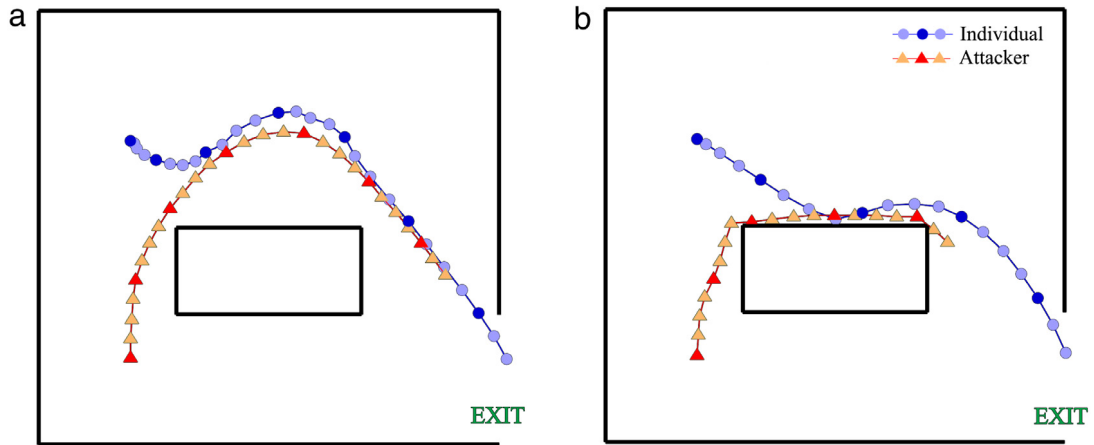


Fig. 8. Simulations based on (a) the driving-forces model and (b) the advanced ESII model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

force model is 176.6 N, giving an acceleration of 2.2 m/s^2 to the individual. This is more reasonable, showing the advantages of the proposed driving force model.

6. Conclusion and future research directions

The agent-based simulation method is often used to do the risk analysis of the crowd under different emergency situations. The individual behavior models, as the foundation of the ABS simulation, are studied for decades. However, the studied situations mostly focus on normal walking or general evacuation. The influences of the threat agents to the individual, especially the moving threat agents, are seldom studied but commonly found in some emergency situations, such as terrorist attacks. At the same time, individual models in special situations need more validations from cases or experiments.

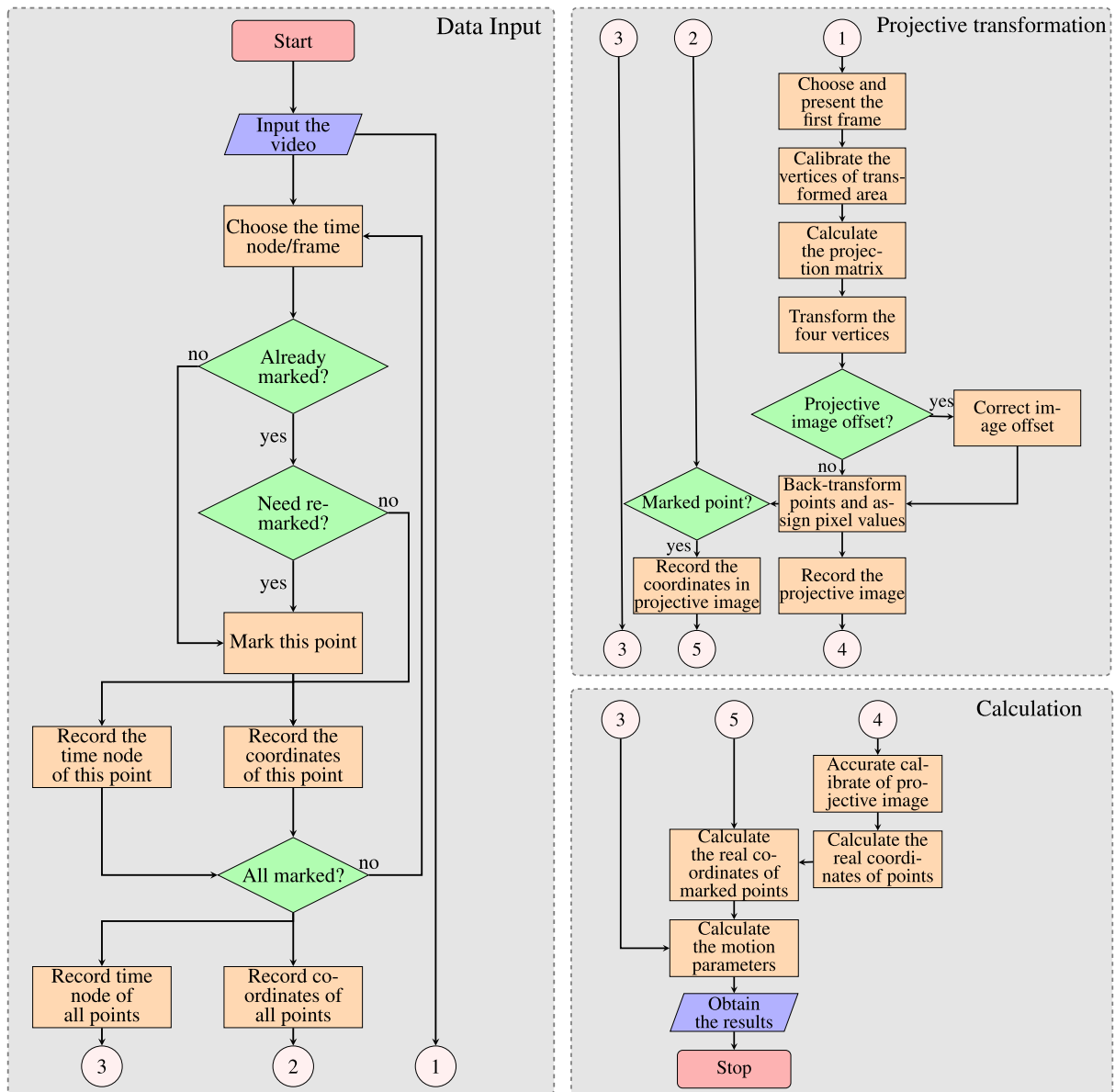


Fig. 9. The flowchart of data process programming.

In this paper, a new individual behavior model in scenarios including moving threat agents is built based on seven individual behavior hypotheses. The model considers not only the repulsive forces caused by other pedestrians or walls, but also the instinctive driving force and the strategic driving force caused by the threat agents. The instinctive driving force is more timely but less rational, with an offset angle away from the threat agents' direction when there is no certain exit in the individual's field of view. On the other hand, the strategic driving force is more tactical and is generated within the decision process. This incorporates the influences of both the threat agents and the exit. At the same time, inner resistance is considered to keep an individual from utter exhaustion. An experiment was conducted to calibrate the key components of the model. The fitting errors between the experimental and simulated trajectories of the driving-forces model is compared to those of an advanced ESII SF model. The results show that the driving-forces model reduced the fitting error by an average of 33.90% and the standard deviation by an average of 44.50%. This result indicates that the driving-forces model is more accurate and stable when doing the simulation in the studied situations. Then the two models are applied to simulate individual behavior in the circumstance that includes an attacker. The simulation results show the practicality of the driving-forces model. Based on these factors, the driving-forces model can be used for simulations in the studied scenarios. For example, the terrorist attacks in public places caused by attackers with close-range weapons.

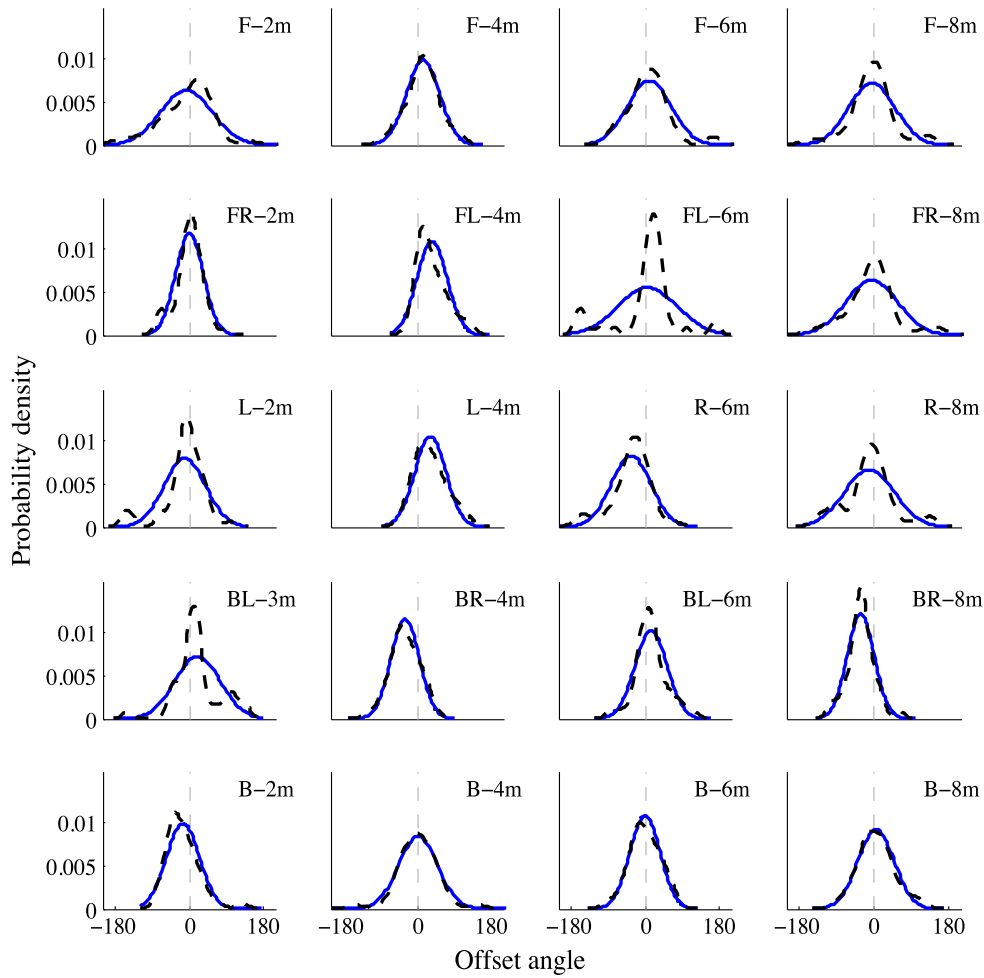


Fig. 10. Probability density curves of the offset angle λ_i (black dotted line stands for the original probability density curve, blue line stands for the fitted normal distribution curve). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

This paper moves forward by building a more reasonable and practical individual behavior model in scenarios including moving threat agents, and getting useful information with real experimental data. From the new model and results, we provide some suggestions to the individual when facing the moving threat agent. First, low level of panic could reduce the unreasonable judgment caused by the instinctive driving force. So we suggest an individual be prepared by previewing the attack scenes in mind when entering a public place. This could help an individual to stay calm when a real attack happens. Second, the direction of the selected exit is an important factor to generate the strategic driving force. So we suggest an individual have a better acknowledgment about the layout of the place where he stays. Third, it takes about 1.5 s to form a behavior strategy by the strategic driving force. So we suggest an individual give himself one or two seconds before taking the evacuation behavior if possible.

The single-exit driving-forces model can also be extended to a multi-exit one when considering the process of choosing the exit to evacuate as studied in Hao et al. [24]. At the same time, the driving-forces model, as well as many other individual behavior models, only considers certain information received through the individual's visual perception. Individuals in the crowd could also be influenced by some kinds of auditory information, for instance, a signal for help or a rumor. At the same time, different kinds of information may have different levels of credibility. A classic case is 2005 Al-Aimmah bridge stampede [25], which is much more complex and cannot be simulated reliably using existing behavior models. A future research direction would include the individual's behavior when receiving different types of uncertain information.

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Appendix A. The flowchart of the data process programming

The flowchart of video data process programming is shown in Fig. 9.

Appendix B. Probability density of offset angle

Probability density curves of offset angle in twenty sets of attacker starting points are shown in Fig. 10.

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